

Module 2 - Series expansion and multivariable calculus

2.1 Taylor's and Maclaurin's series for one variable

Introduction:

Taylor's series: $f(x) = f(a) + \frac{x-a}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots$

Maclaurin's series: $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

Maclaurin's series is a Taylor's series expansion of a function at the origin.

Problems:

1. Obtain the Maclaurin's series expansion for the following functions:

$$\sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} + \dots \quad (\text{May 22})$$

$$f(x) = \sqrt{1 + \sin 2x} = \sqrt{1 + 2\sin x \cos x} = \sqrt{\sin^2 x + \cos^2 x + 2\sin x \cos x} = \sqrt{(\sin x + \cos x)^2}$$

$y = \sin x + \cos x$	$f(0) = 1$
$y_1 = \cos x - \sin x$	$f'(0) = 1$
$y_2 = -\sin x - \cos x = -y$	$f''(0) = -1$
$y_3 = -y_1$	$f'''(0) = -1$
$y_4 = -y_2$	$f^{iv}(0) = 1$
$y_5 = -y_3$	$f^v(0) = 1$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots$$

$$\sqrt{1 + \sin 2x} = 1 + \frac{x}{1!}(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(1) + \dots$$

$$\Rightarrow \sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

2. $\sec x$ upto x^4 term

$$y = \sec x$$

$$y_1 = \sec x \tan x = y \tan x$$

$$\begin{aligned} y_2 &= y \sec^2 x + y_1 \tan x = y^3 + y \tan^2 x \\ &= y^3 + y(\sec^2 x - 1) = y^3 + y^3 - y \\ &= 2y^3 - y \end{aligned}$$

$$y_3 = 6y^2 y_1 - y_1$$

$$y_4 = 12y y_1^2 + 6y^2 y_2 - y_2$$

$$f(0) = 1$$

$$f'(0) = 0$$

$$f''(0) = 1$$

$$f'''(0) = 0$$

$$f^{iv}(0) = 5$$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \dots$$

$$\sec x = 1 + 0 + \frac{x^2}{2!} (1) + 0 + \frac{x^4}{4!} (5) + \dots = 1 + \frac{x^2}{2} + \frac{5x^4}{24}$$

3. $\log(1 + e^x)$ up to 3rd degree term.

$$y = \log(1 + e^x)$$

$$y_1 = \frac{e^x}{1+e^x}$$

$$y_2 = \frac{(1+e^x)e^x - e^x e^x}{(1+e^x)^2} = y_1 - y_1^2$$

$$y_3 = y_2 - 2y_1 y_2$$

$$y_4 = y_3 - 2(y_1 y_3 + y_2^2)$$

$$f(0) = \log 2$$

$$f'(0) = \frac{1}{2}$$

$$f''(0) = \frac{1}{4}$$

$$f'''(0) = 0$$

$$f^{iv}(0) = -\frac{1}{8}$$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \dots$$

$$\begin{aligned} \log(1 + e^x) &= \log 2 + \frac{x}{1!} \left(\frac{1}{2}\right) + \frac{x^2}{2!} \left(\frac{1}{4}\right) + \frac{x^3}{3!} (0) + \frac{x^4}{4!} \left(-\frac{1}{8}\right) \\ &= \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} \end{aligned}$$

4. **Prove that** $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

$y = \log(1+x)$	$f(0) = 0$
$y_1 = \frac{1}{1+x}$	$f'(0) = 1$
$y_2 = -\frac{1}{(1+x)^2} = -y_1^2$	$f''(0) = -1$
$y_3 = -2y_1y_2$	$f'''(0) = 2$
$y_4 = -2y_1y_3 - 2y_2^2$	$f^{iv}(0) = -6$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{iv}(0) + \dots$$

$$\begin{aligned}\log(1+x) &= 0 + \frac{x}{1!} - \frac{x^2}{2!} + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(-6) + \dots \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots\end{aligned}$$

5. **tan x = x + $\frac{2x^3}{3!} + \frac{16x^5}{5!} + \dots$**

$y = \tan x$	$y(0) = 0$
$y_1 = \sec^2 x = 1 + \tan^2 x = 1 + y^2$	$y_1(0) = 1$
$y_2 = 2yy_1 = 2y + 2y^3$	$y_2(0) = 0$
$y_3 = 2y_1^2 + 2yy_2$	$y_3(0) = 2$
$y_4 = 4y_1y_2 + 2y_1y_2 + 2yy_3 = 6y_1y_2 + 2yy_3$	$y_4(0) = 0$
$y_5 = 6y_2^2 + 6y_1y_3 + 2y_1y_3 + 2yy_4$	$y_5(0) = 16$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

$$\begin{aligned}\tan x &= 0 + x + 0 + \frac{x^3}{3} + 0 + \frac{x^5}{5!}(16) + \dots \\ &= x + \frac{x^3}{3} + \frac{x^5}{5!}(16) + \dots\end{aligned}$$

$$6. \log \sec x = \frac{x^2}{2!} + \frac{2x^4}{4!} + \frac{16x^6}{6!} + \dots$$

$$y = \log \sec x$$

$$y_1 = \frac{1}{\sec x} \sec x \tan x = \tan x$$

$$y_2 = \sec^2 x = 1 + \tan^2 x = 1 + y_1^2$$

$$y_3 = 2y_1y_2$$

$$y_4 = 2y_2^2 + 2y_1y_3$$

$$y_5 = 4y_2y_3 + 2y_1y_4 + 2y_2y_3 = 6y_2y_3 + 2y_1y_4$$

$$y_6 = 6y_2y_4 + 6y_3^2 + 2y_1y_5 + 2y_2y_4$$

$$y(0) = 0$$

$$y_1(0) = 0$$

$$y_2(0) = 1$$

$$y_3(0) = 0$$

$$y_4(0) = 2$$

$$y_5(0) = 0$$

$$y_6(0) = 16$$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0) + \frac{x^5}{5!}f^{(5)}(0) + \dots$$

$$\log \sec x = 0 + 0 + 0 + \frac{x^2}{2!} + 0 + \frac{x^4}{4!}(2) + 0 + \frac{x^6}{6!}(16) + \dots$$

$$= \frac{x^2}{2!} + \frac{2x^4}{4!} + \frac{x^6}{6!}(16) + \dots$$

Home work:

$$7. \frac{e^x}{1+e^x} \text{ up to 3rd degree term.}$$

$$\text{Hint: } y = \frac{e^x}{1+e^x}, y_1 = \frac{(1+e^x)e^x - e^{2x}}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2} = y_1^2$$

$$\text{Ans: } \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \dots$$

$$8. e^{\sin x} = 1 + x + \frac{x^2}{2!} - \frac{3x^4}{4!} - \frac{8x^5}{5!} + \dots$$

$$\text{Hint: } y = 1, y_1 = 1, y_2 = 1, y_3 = 0, y_4 = -3, y_5 = -8$$

$$9. e^{\tan^{-1} x} = 1 + x + \frac{x^2}{2} - \frac{x^3}{3!} - \frac{7x^4}{4!} + \dots$$

$$\text{Hint: } y = 1, y_1 = 1, y_2 = 1, y_3 = -1, y_4 = -7, y_5 = 5$$

$$10. \log(1 + e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$$

$$\text{Hint: } y = \log 2, y_1 = \frac{1}{2}, y_2 = \frac{1}{4}, y_3 = 0, y_4 = -\frac{1}{8}$$

2.2 Evaluation of indeterminate forms

Basic results:

$$\diamond \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

$$\diamond \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

$$\diamond \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

Introduction:

- ❖ Indeterminate forms: $\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 0^0, 1^\infty, \infty^0$.
- ❖ Limits which lead to indeterminate forms are evaluated by using L' Hospital's rule.
- ❖ L' Hospital's rule: Suppose $f(x)$ and $g(x)$ are functions such that
(i) $\lim_{x \rightarrow a} f(x) = 0, \lim_{x \rightarrow a} g(x) = 0$ (ii) $f'(x)$ and $g'(x)$ exist and $g'(a) \neq 0$,

Then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ provided the limit on the RHS exists.

1. Prove that $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$

$$\text{Let } L = \lim_{x \rightarrow 0} (1+x)^{1/x}$$

$$\log L = \lim_{x \rightarrow 0} \log(1+x)^{1/x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{x} \log(1+x)$$

$$= \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \lim_{x \rightarrow 0} \frac{1}{1+x} = 1$$

By taking anti log, $L = e$

2. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x}{3} \right)^{\frac{1}{x}}$

$$\text{Let } L = \lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x}{3} \right)^{\frac{1}{x}}$$

$$\begin{aligned} \log L &= \lim_{x \rightarrow 0} \log \left(\frac{1^x + 2^x + 3^x}{3} \right)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \log \frac{1^x + 2^x + 3^x}{3} \\ &= \lim_{x \rightarrow 0} \frac{\log(1^x + 2^x + 3^x) - \log 3}{x} = \frac{0}{0} \text{ form} \end{aligned}$$

By L.H Rule,

$$\begin{aligned} \log L &= \lim_{x \rightarrow 0} \frac{1}{1^x + 2^x + 3^x} (1^x \log 1 + 2^x \log 2 + 3^x \log 3) \\ &= \frac{1}{3} (\log 1 + \log 2 + \log 3) \\ &= \log 6^{\frac{1}{3}} \end{aligned}$$

By taking anti log, $L = 6^{\frac{1}{3}}$

3. Prove that $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{\frac{1}{x}} = (abc)^{\frac{1}{3}}$ (May 22)

$$\text{Let } L = \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{\frac{1}{x}}$$

$$\begin{aligned} \log L &= \lim_{x \rightarrow 0} \log \left(\frac{a^x + b^x + c^x}{3} \right)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \log \frac{a^x + b^x + c^x}{3} \\ &= \lim_{x \rightarrow 0} \frac{\log(a^x + b^x + c^x) - \log 3}{x} = \frac{0}{0} \text{ form} \end{aligned}$$

By L.H Rule,

$$\begin{aligned} \log L &= \lim_{x \rightarrow 0} \frac{1}{a^x + b^x + c^x} (a^x \log a + b^x \log b + c^x \log c) \\ &= \frac{1}{3} (\log a + \log b + \log c) \\ &= \log(abc)^{\frac{1}{3}} \end{aligned}$$

Taking anti log, $L = (abc)^{\frac{1}{3}}$

4. Prove that $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} = e^{\frac{1}{3}}$

$$\text{Let } L = \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}}$$

$$\log L = \lim_{x \rightarrow 0} \log \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \frac{\tan x}{x} = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \lim_{x \rightarrow 0} \frac{1}{2x} \cdot \frac{x}{\tan x} \left(\frac{x \sec^2 x - \tan x}{x^2} \right)$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{x \sec^2 x - \tan x}{x^3} \right) = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sec^2 x + x \cdot 2 \sec x \cdot \sec x \tan x - \sec^2 x}{3x^2} \right)$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{2 \sec x \cdot \sec x \tan x}{3x} \right)$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right) \sec^2 x = \frac{1}{3}$$

By taking anti log, $L = e^{\frac{1}{3}}$

5. Prove that $\lim_{x \rightarrow \frac{\pi}{4}} (\tan x)^{\tan 2x} = \frac{1}{e}$

$$\text{Let } L = \lim_{x \rightarrow \frac{\pi}{4}} (\tan x)^{\tan 2x}$$

$$\log L = \lim_{x \rightarrow \frac{\pi}{4}} \log(\tan x)^{\tan 2x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \tan 2x \log \tan x$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\log \tan x}{\cot 2x} = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\frac{1}{\tan x} \sec^2 x}{-2 \operatorname{cosec}^2 2x} \right)$$

$$\log L = -\lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\sin^2 2x}{2 \sin x \cos x} \right)$$

$$= -\lim_{x \rightarrow \frac{\pi}{4}} \sin 2x = -1$$

$$\text{Taking anti log, } L = e^{-1} = \frac{1}{e}$$

6. Prove that $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)} = e^{\frac{2}{\pi}}$

$$\text{Let } L = \lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$$

$$\log L = \lim_{x \rightarrow a} \log \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$$

$$= \lim_{x \rightarrow a} \tan\left(\frac{\pi x}{2a}\right) \log \left(2 - \frac{x}{a} \right)$$

$$= \lim_{x \rightarrow a} \frac{\log\left(2 - \frac{x}{a}\right)}{\cot\left(\frac{\pi x}{2a}\right)} = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \lim_{x \rightarrow a} \frac{\left(-\frac{1}{a}\right) \frac{1}{2 - \frac{x}{a}}}{\left(-\frac{\pi}{2a}\right) \operatorname{cosec}^2 \frac{\pi x}{2a}}$$

$$\log L = \frac{2}{\pi} \lim_{x \rightarrow a} \frac{\sin^2 \frac{\pi x}{2a}}{2 - \frac{x}{a}} = \frac{2}{\pi}$$

Taking anti log, $L = e^{\frac{2}{\pi}}$

7. **Prove that $\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} = 1$**

$$\text{Let } L = \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$$

$$\log L = \lim_{x \rightarrow \frac{\pi}{2}} \log(\sin x)^{\tan x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \tan x \log \sin x$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\log \sin x}{\cot x} = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x}{-\operatorname{cosec}^2 x} = 0$$

Taking anti log, $L = e^0 = 1$

8. **Prove that $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^x = 1$**

$$\text{Let } L = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^x$$

$$\log L = \lim_{x \rightarrow \infty} \log \left(1 + \frac{1}{x^2}\right)^x$$

$$= \lim_{x \rightarrow \infty} x \log \left(1 + \frac{1}{x^2}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{\log \left(1 + \frac{1}{x^2}\right)}{\frac{1}{x}} = \frac{0}{0} \text{ form}$$

By L.H rule,

$$\log L = \lim_{x \rightarrow \infty} \frac{\frac{1}{1 + \frac{1}{x^2}} \left(-\frac{2}{x^3}\right)}{\left(-\frac{1}{x^2}\right)} = 2 \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1 + \frac{1}{x^2}} = 0$$

Taking anti log, $L = e^0 = 1$.

Home work:

9. Prove that $\lim_{x \rightarrow 0} (a^x + x)^{\frac{1}{x}} = ae$

Hint: $\log L = \lim_{x \rightarrow 0} \frac{\log(a^x + x)}{x} = \frac{0}{0}$

$$\log L = \lim_{x \rightarrow 0} \frac{a^x \log a + 1}{a^x + x} = \log a + 1 = \log a + \log e = \log ae$$

10. Prove that $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x + d^x}{4} \right)^{\frac{1}{x}} = (abcd)^{\frac{1}{4}}$

11. Prove that $\lim_{x \rightarrow 0} (\cot x)^{\frac{1}{\log x}} = \frac{1}{e}$

12. Prove that $\lim_{x \rightarrow 1} (1 - x^2)^{\frac{1}{\log(1-x)}} = e$

13. Prove that $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}} = e^{-\left(\frac{1}{6}\right)}$

14. Prove that $\lim_{x \rightarrow 0} (a^x + x)^{\frac{1}{x}} = ae$

15. Prove that $\lim_{x \rightarrow \frac{\pi}{2}} (\cos x)^{\frac{\pi}{2} - x} = 0$

2.4 Partial differentiation

Introduction:

Let $f(x, y)$ be a function of two independent variables x and y .

- ❖ **First order partial derivatives:** $f_x = \frac{\partial f}{\partial x}, f_y = \frac{\partial f}{\partial y}$
- ❖ **Second order partial Derivatives:** $f_{xx} = \frac{\partial^2 f}{\partial x^2}, f_{yy} = \frac{\partial^2 f}{\partial y^2}, f_{xy} = \frac{\partial^2 f}{\partial x \partial y}, f_{yx} = \frac{\partial^2 f}{\partial y \partial x}$
- ❖ **Property:** $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

Problems:

1. If $z = 4x^2 + 8x^3y^2 + 6xy^2 + 8y + 6$. Find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

Solution:

$$z = 4x^2 + 8x^3y^2 + 6xy^2 + 8y + 6 \text{ ---- (1)}$$

Differentiate (1) partially w. r. to x

$$\frac{\partial z}{\partial x} = 8x + 24x^2y^2 + 6y^2$$

Differentiate (1) partially w. r. to y

$$\frac{\partial z}{\partial y} = 16x^3y + 12xy + 8.$$

2. If $z = f(x + ct) + g(x - ct)$, Prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$ (MQP)

Solution:

$$z = f(x + ct) + g(x - ct) \text{ ---- (1)}$$

Differentiate (1) partially w. r. to x

$$\frac{\partial z}{\partial x} = f'(x + ct) + g'(x - ct)$$

Differentiate (1) again partially w. r. to x

$$\frac{\partial^2 z}{\partial x^2} = f''(x + ct) + g''(x - ct)$$

Differentiate (1) partially w. r. to t

$$\frac{\partial z}{\partial t} = cf'(x + ct) - cg'(x - ct)$$

Differentiate (1) again partially w. r. to t

$$\begin{aligned} \frac{\partial^2 z}{\partial t^2} &= c^2 f''(x + ct) + c^2 g''(x - ct) \\ &= c^2 [f''(x + ct) + g''(x - ct)] \\ &= c^2 \frac{\partial^2 z}{\partial x^2} \end{aligned}$$

Therefore,

$$\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$$

3. If $z = e^{ax+by}f(ax-by)$ Prove that $b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz$ (May 22)

$$z = e^{ax+by}f(ax-by) \text{ ---- (1)}$$

Differentiate (1) partially w. r. to x

$$\begin{aligned}\frac{\partial z}{\partial x} &= ae^{ax+by}f'(ax-by) + ae^{ax+by}f(ax-by) \\ &= ae^{ax+by}f'(ax-by) + az\end{aligned}$$

Differentiate (1) partially w. r. to y

$$\begin{aligned}\frac{\partial z}{\partial y} &= -be^{ax+by}f'(ax-by) + be^{ax+by}f(ax-by) \\ &= -be^{ax+by}f'(ax-by) + bz\end{aligned}$$

Therefore,

$$b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = abe^{ax+by}f'(ax-by) + abz - abe^{ax+by}f'(ax-by) + abz$$

$$\therefore b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz$$

Note: $(x+y+z)(x^2+y^2+z^2-xy-yz-zx) = x^3+y^3+z^3-3xyz$

4. If $u = \log(x^3+y^3+z^3-3xyz)$. then Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

and hence show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

$$u = \log(x^3+y^3+z^3-3xyz)$$

$$\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial y} = \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$$

Case 1:

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz} + \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz} + \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz} \\ &= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{x^3+y^3+z^3-3xyz} \\ &= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)} \\ &= \frac{3}{x+y+z}\end{aligned}$$

Case 2:

$$\begin{aligned}\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) u \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}\right) \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{3}{x+y+z}\right) = \frac{-9}{(x+y+z)^2}\end{aligned}$$

5. If $z(x + y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$

$$z(x + y) = x^2 + y^2 \text{ ---- (1)}$$

Differentiate (1) partially w. r. to x

$$\frac{\partial z}{\partial x}(x + y) + z(1 + 0) = 2x + 0$$

$$\frac{\partial z}{\partial x} = \frac{2x - z}{x + y}$$

Differentiate (1) partially w. r. to y

$$\frac{\partial z}{\partial y}(x + y) + z(0 + 1) = 0 + 2y$$

$$\frac{\partial z}{\partial y} = \frac{2y - z}{x + y}$$

Therefore,

$$\begin{aligned} \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 &= \left(\frac{2x - z}{x + y} - \frac{2y - z}{x + y}\right)^2 \\ &= \left(\frac{2x - z - 2y + z}{x + y}\right)^2 \\ &= 4\left(\frac{x - y}{x + y}\right)^2 \text{ ----- (2)} \end{aligned}$$

$$\begin{aligned} 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) &= 4\left(1 - \frac{2x - z}{x + y} - \frac{2y - z}{x + y}\right) \\ &= \frac{4}{x + y}(x + y - 2x + z - 2y + z) \\ &= \frac{4}{x + y}(2z - x - y) \\ &= \frac{4}{x + y}\left[2\left(\frac{x^2 + y^2}{x + y}\right) - (x + y)\right] \\ &= \frac{4}{(x + y)^2}(2x^2 + 2y^2 - x^2 - y^2 - 2xy) \\ &= 4\left(\frac{x^2 + y^2 - 2xy}{(x + y)^2}\right) = 4\left(\frac{x - y}{x + y}\right)^2 \text{ ----- (3)} \end{aligned}$$

Equating (2) and (3),

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$$

6. If $v = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$ prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$

$$(x^2 + y^2 + z^2)^{\frac{1}{2}} = 1/v$$

$$\frac{1}{v^2} = x^2 + y^2 + z^2$$

Differentiate partially w. r. to x

$$-\frac{2}{v^3} \frac{\partial v}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = -xv^3$$

Differentiate again partially w. r. to x

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} &= -v^3 - 3xv^2 \frac{\partial v}{\partial x} \\ &= -v^3 - 3xv^2(-xv^3) \\ &= -v^3 + 3x^2v^5\end{aligned}$$

Similarly,

$$\frac{\partial^2 v}{\partial y^2} = -v^3 + 3y^2v^5$$

$$\frac{\partial^2 v}{\partial z^2} = -v^3 + 3z^2v^5$$

Therefore,

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} &= -3v^3 + 3(x^2 + y^2 + z^2)v^5 \\ &= -3v^3 + 3\frac{v^5}{v^2} \\ &= -3v^3 + 3v^3 = 0\end{aligned}$$

Home work:

7. If $u = x^2 \tan^{-1} \frac{y}{x} - y^2 \tan^{-1} \frac{x}{y}$, prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$.

8. If $u = \tan^{-1} \left(\frac{2xy}{x^2 - y^2} \right)$ prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

9. Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$

10. If $u = x^y$ show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$

2.5 Total differentiation

Introduction:

❖ If $u = u(x, y)$ where $x = x(t)$ and $y = y(t)$ then $\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$

❖ If $u = u(x, y, z)$ where $x = x(t)$, $y = y(t)$ and $z = z(t)$ then

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

Problems:

1. If $z = u^2 + v^2$ where $u = at^2$ and $v = 2at$ find $\frac{dz}{dt}$.

$$z = u^2 + v^2$$

$$\frac{\partial z}{\partial u} = 2u, \quad \frac{\partial z}{\partial v} = 2v$$

$u = at^2$	$v = 2at$
$\frac{du}{dt} = 2at = v$	$\frac{dv}{dt} = 2a$

Therefore,

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}$$

$$= (2u)v + (2v)2a$$

$$= 2at^2(2at) + 4at(2a)$$

$$= 4a^2t(t^2 + 2)$$

2. If $u = xy^2 + x^2y$ with $x = at^2, y = 2at$ find $\frac{du}{dt}$ using partial derivatives.

$$u = xy^2 + x^2y$$

$$\frac{\partial u}{\partial x} = y^2 + 2xy, \quad \frac{\partial u}{\partial y} = 2xy + x^2$$

$x = at^2$	$y = 2at$
$\frac{dx}{dt} = 2at = y$	$\frac{dy}{dt} = 2a$

$$\text{Therefore, } \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$\begin{aligned} \frac{du}{dt} &= (y^2 + 2xy)y + (2xy + x^2)2a \\ &= y^3 + 2xy^2 + 4axy + 2ax^2 \\ &= (2at)^3 + 2(at^2)(2at)^2 + 4a(at^2)(2at) + 2a(at^2)^2 \\ &= 2a^3t^3(4 + 4t + 4 + t) \\ &= 2a^3t^3(8 + 5t) \end{aligned}$$

3. If $u = \tan^{-1}\left(\frac{y}{x}\right)$ where $x = e^t - e^{-t}$ and $y = e^t + e^{-t}$ find $\frac{du}{dt}$

$$u = \tan^{-1} \frac{y}{x}$$

$$\frac{\partial u}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(\frac{1}{x}\right) = \frac{x}{x^2 + y^2}$$

$x = e^t - e^{-t}$	$y = e^t + e^{-t}$
$\frac{dx}{dt} = e^t + e^{-t} = y$	$\frac{dy}{dt} = e^t - e^{-t} = x$

$$\begin{aligned} \text{Therefore, } \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= \left(-\frac{y}{x^2 + y^2}\right)y + \left(\frac{x}{x^2 + y^2}\right)x \\ &= \frac{x^2 - y^2}{x^2 + y^2} \\ &= \frac{(e^t - e^{-t})^2 - (e^t + e^{-t})^2}{(e^t - e^{-t})^2 + (e^t + e^{-t})^2} = \frac{-2}{e^{2t} + e^{-2t}} \end{aligned}$$

4. If $u = x^2 + y^2 + z^2$, where $x = e^{2t}$, $y = e^{2t} \cos 3t$, $z = e^{2t} \sin 3t$ find $\frac{du}{dt}$.

$$u = x^2 + y^2 + z^2$$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = 2y, \quad \frac{\partial u}{\partial z} = 2z$$

$x = e^{2t}$	$y = e^{2t} \cos 3t$	$z = e^{2t} \sin 3t$
$\frac{dx}{dt} = 2e^{2t}$	$\frac{dy}{dt} = 2e^{2t} \cos 3t - 3e^{2t} \sin 3t$	$\frac{dz}{dt} = 2e^{2t} \sin 3t + 3e^{2t} \cos 3t$
$= 2x$	$= 2y - 3z$	$= 2z + 3y$

Therefore,

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt} \\ &= 2x(2x) + 2y(2y - 3z) + 2z(2z + 3y) \\ &= 4(x^2 + y^2 + z^2) \\ &= 4(e^{4t} + e^{4t}) \\ &= 8e^{4t} \end{aligned}$$

5. If $u = e^x \sin(yz)$, where $x = t^2$, $y = t - 1$, $z = \frac{1}{t}$ find $\frac{du}{dt}$ at $t = 1$.

$$u = e^x \sin(yz)$$

$$\frac{\partial u}{\partial x} = e^x \sin(yz), \quad \frac{\partial u}{\partial y} = ze^x \cos(yz), \quad \frac{\partial u}{\partial z} = ye^x \cos(yz)$$

$x = t^2$	$y = t - 1$	$z = \frac{1}{t}$
$\frac{dx}{dt} = 2t$	$\frac{dy}{dt} = 1$	$\frac{dz}{dt} = -\frac{1}{t^2}$

At $t = 1$,

$$x = 1, \quad y = 0, \quad z = 1.$$

$$\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = -1$$

$$\frac{\partial u}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = e, \quad \frac{\partial u}{\partial z} = 0$$

Therefore,

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt} \\ &= 0 + e \cdot 1 + 0 \\ &= e \end{aligned}$$

6. If $u = x^2 + xy + y^2, x = t^2, y = 3t$ then find $\frac{du}{dt}$.

$$u = x^2 + xy + y^2$$

$$\frac{\partial u}{\partial x} = 2x + y, \quad \frac{\partial u}{\partial y} = x + 2y$$

$x = t^2$	$y = 3t$
$\frac{dx}{dt} = 2t$	$\frac{dy}{dt} = 3$

Therefore,

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= (2x + y)2t + (x + 2y)3 \\ &= (2t^2 + 3t)2t + (t^2 + 6t)3 \\ &= 4t^3 + 6t^2 + 3t^2 + 18t \\ &= 4t^3 + 9t^2 + 18t \end{aligned}$$

7. If $u = x^3y^2 + x^2y^3$ with $x = at^2, y = 2at$ find $\frac{du}{dt}$ using partial derivatives.

$$u = x^3y^2 + x^2y^3$$

$$\frac{\partial u}{\partial x} = 3x^2y^2 + 2xy^3, \quad \frac{\partial u}{\partial y} = 2x^3y + 3x^2y^2$$

$x = at^2$	$y = 2at$
$\frac{dx}{dt} = 2at = y$	$\frac{dy}{dt} = 2a$

Therefore,

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= (3x^2y^2 + 2xy^3)y + (2x^3y + 3x^2y^2)2a \\ &= 3x^2y^3 + 2xy^4 + 4ax^3y + 6ax^2y^2 \\ &= 3a^2t^4 \cdot 8a^3t^3 + 2at^2 \cdot 16a^4t^4 + 4a \cdot a^3t^6 \cdot 2at + 6a \cdot a^2t^4 \cdot 4a^2t^2 \\ &= 8a^5t^6(3t + 4 + t + 3) \\ &= 8a^5t^6(4t + 7) \end{aligned}$$

8. If $u = \sin \frac{x}{y}$, where $x = e^t, y = e^{t^2}$ find $\frac{du}{dt}$.

$$u = \sin \frac{x}{y}$$

$$\frac{\partial u}{\partial x} = \frac{1}{y} \cos \frac{x}{y}, \quad \frac{\partial u}{\partial y} = -\frac{x}{y^2} \cos \frac{x}{y}$$

$$x = e^t, \quad \frac{dx}{dt} = e^t = x$$

$$y = e^{t^2}, \quad \frac{dy}{dt} = 2te^{t^2} = 2ty$$

Therefore,

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= \left(\frac{1}{y} \cos \frac{x}{y} \right) x + \left(-\frac{x}{y^2} \cos \frac{x}{y} \right) 2ty \\ &= (1 - 2t) \frac{x}{y} \cos \frac{x}{y} \\ &= (1 - 2t) (e^{t-t^2} \cos e^{t-t^2}) \end{aligned}$$

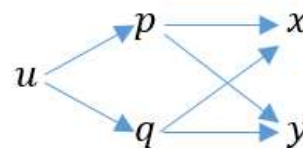
2.6 Partial derivatives of composite functions

Introduction

- ❖ If $u = f(p, q)$ where $p = p(x, y)$ and $q = q(x, y)$ then z is a composite function of x and y . Partial derivatives of composite function z are given by

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial y}$$

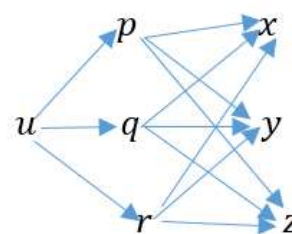


- ❖ If $u = f(p, q, r)$ where $p = p(x, y)$, $q = q(x, y)$ and $r = r(x, y)$ then z is a composite function of x and y . Partial derivatives of composite function z are given by

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x},$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial y} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial z} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial z}$$



Problems:

1. If $z = f(x, y)$, where $x = e^u + e^{-v}$, $y = e^{-u} - e^v$ then P.T. $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$.

$x = e^u + e^{-v}$	$y = e^{-u} - e^v$
$\frac{\partial x}{\partial u} = e^u$	$\frac{\partial y}{\partial u} = -e^{-u}$
$\frac{\partial x}{\partial v} = -e^{-v}$	$\frac{\partial y}{\partial v} = -e^v$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} = \frac{\partial z}{\partial x} (e^u) + \frac{\partial z}{\partial y} (-e^{-u}) \quad \text{----- (1)}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} = \frac{\partial z}{\partial x} (-e^{-v}) + \frac{\partial z}{\partial y} (-e^v) \quad \text{----- (2)}$$

(1) – (2) gives,

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} (e^u + e^{-v}) + \frac{\partial z}{\partial y} (-e^{-u} + e^v)$$

$$= (e^u + e^{-v}) \frac{\partial z}{\partial x} - (e^{-u} - e^v) \frac{\partial z}{\partial y}$$

$$= x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$$

2. If $z = f(x, y)$, where $x = e^u \cos v$, $y = e^u \sin v$ then P.T. $y \frac{\partial z}{\partial u} + x \frac{\partial z}{\partial v} = e^{2u} \frac{\partial z}{\partial y}$.

$x = e^u \cos v$	$y = e^u \sin v$
$\frac{\partial x}{\partial u} = e^u \cos v = x$	$\frac{\partial y}{\partial u} = e^u \sin v = y$
$\frac{\partial x}{\partial v} = -e^u \sin v = -y$	$\frac{\partial y}{\partial v} = e^u \cos v = x$

$$\begin{aligned}
 y \frac{\partial z}{\partial u} &= y \left\{ \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \right\} \\
 &= y \left\{ \frac{\partial z}{\partial x} (x) + \frac{\partial z}{\partial y} (y) \right\} \\
 &= xy \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} \quad \text{---- (1)}
 \end{aligned}$$

$$\begin{aligned}
 x \frac{\partial z}{\partial v} &= x \left\{ \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} \right\} \\
 &= x \left\{ \frac{\partial z}{\partial x} (-y) + \frac{\partial z}{\partial y} (x) \right\} \\
 &= -xy \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} \quad \text{---- (2)}
 \end{aligned}$$

(1) + (2) gives,

$$\begin{aligned}
 y \frac{\partial z}{\partial u} + x \frac{\partial z}{\partial v} &= xy \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} - xy \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} \\
 &= (x^2 + y^2) \frac{\partial z}{\partial y} \\
 &= (e^{2u} \cos^2 u + e^{2u} \sin^2 v) \frac{\partial z}{\partial y} \\
 &= e^{2u} \frac{\partial z}{\partial y}
 \end{aligned}$$

3. If $u = f(x - y, y - z, z - x)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

$p = x - y$	$q = y - z$	$r = z - x$
$\frac{\partial p}{\partial x} = 1$	$\frac{\partial q}{\partial x} = 0$	$\frac{\partial r}{\partial x} = -1$
$\frac{\partial p}{\partial y} = -1$	$\frac{\partial q}{\partial y} = 1$	$\frac{\partial r}{\partial y} = 0$
$\frac{\partial p}{\partial z} = 0$	$\frac{\partial q}{\partial z} = -1$	$\frac{\partial r}{\partial z} = 1$

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} \\ &= \frac{\partial u}{\partial p} (1) + \frac{\partial u}{\partial q} (0) + \frac{\partial u}{\partial r} (-1) \\ &= \frac{\partial u}{\partial p} - \frac{\partial u}{\partial r} \quad \text{---- (1)}\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial u}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial y} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} \\ &= \frac{\partial u}{\partial p} (-1) + \frac{\partial u}{\partial q} (1) + \frac{\partial u}{\partial r} (0) \\ &= \frac{\partial u}{\partial q} - \frac{\partial u}{\partial p} \quad \text{---- (2)}\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial z} &= \frac{\partial u}{\partial p} \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial z} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} \\ &= \frac{\partial u}{\partial p} (0) + \frac{\partial u}{\partial q} (-1) + \frac{\partial u}{\partial r} (1) \\ &= \frac{\partial u}{\partial r} - \frac{\partial u}{\partial q} \quad \text{---- (3)}\end{aligned}$$

(1) + (2) + (3) gives,

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial p} - \frac{\partial u}{\partial q} + \frac{\partial u}{\partial q} - \frac{\partial u}{\partial r} + \frac{\partial u}{\partial r} - \frac{\partial u}{\partial p} \\ &= 0\end{aligned}$$

4. If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$ then prove that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$.

$p = 2x - 3y$	$q = 3y - 4z$	$r = 4z - 2x$
$\frac{\partial p}{\partial x} = 2$	$\frac{\partial q}{\partial x} = 0$	$\frac{\partial r}{\partial x} = -2$
$\frac{\partial p}{\partial y} = -3$	$\frac{\partial q}{\partial y} = 3$	$\frac{\partial r}{\partial y} = 0$
$\frac{\partial p}{\partial z} = 0$	$\frac{\partial q}{\partial z} = -4$	$\frac{\partial r}{\partial z} = 4$

Therefore,

$$\begin{aligned} \frac{1}{2} \frac{\partial u}{\partial x} &= \frac{1}{2} \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} \right\} \\ &= \frac{1}{2} \left\{ \frac{\partial u}{\partial p} (2) + 0 + \frac{\partial u}{\partial r} (-2) \right\} \\ &= \frac{\partial u}{\partial p} - \frac{\partial u}{\partial r} \end{aligned}$$

$$\begin{aligned} \frac{1}{3} \frac{\partial u}{\partial y} &= \frac{1}{3} \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial y} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} \right\} \\ &= \frac{1}{3} \left\{ \frac{\partial u}{\partial p} (-3) + \frac{\partial u}{\partial q} (3) + 0 \right\} \\ &= \frac{\partial u}{\partial q} - \frac{\partial u}{\partial p} \end{aligned}$$

$$\begin{aligned} \frac{1}{4} \frac{\partial u}{\partial z} &= \frac{1}{4} \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial z} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} \right\} \\ &= \frac{1}{4} \left\{ 0 + \frac{\partial u}{\partial q} (-4) + \frac{\partial u}{\partial r} (4) \right\} \\ &= \frac{\partial u}{\partial r} - \frac{\partial u}{\partial q} \end{aligned}$$

(1) + (2) + (3) gives,

$$\begin{aligned} \frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial p} - \frac{\partial u}{\partial r} + \frac{\partial u}{\partial q} - \frac{\partial u}{\partial p} + \frac{\partial u}{\partial r} - \frac{\partial u}{\partial q} \\ &= 0 \end{aligned}$$

5. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$. (May 22)

$p = \frac{x}{y}$	$q = \frac{y}{z}$	$r = \frac{z}{x}$
$\frac{\partial p}{\partial x} = \frac{1}{y}$	$\frac{\partial q}{\partial x} = 0$	$\frac{\partial r}{\partial x} = -\frac{z}{x^2}$
$\frac{\partial p}{\partial y} = -\frac{x}{y^2}$	$\frac{\partial q}{\partial y} = \frac{1}{z}$	$\frac{\partial r}{\partial y} = 0$
$\frac{\partial p}{\partial z} = 0$	$\frac{\partial q}{\partial z} = -\frac{y}{z^2}$	$\frac{\partial r}{\partial z} = \frac{1}{x}$

$$\begin{aligned}
 x \frac{\partial u}{\partial x} &= x \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} \right\} \\
 &= x \left\{ \frac{\partial u}{\partial p} \left(\frac{1}{y} \right) + 0 + \frac{\partial u}{\partial r} \left(-\frac{z}{x^2} \right) \right\} \\
 &= \frac{x}{y} \frac{\partial u}{\partial p} - \frac{z}{x} \frac{\partial u}{\partial r} \quad \text{---- (1)}
 \end{aligned}$$

$$\begin{aligned}
 y \frac{\partial u}{\partial y} &= y \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial y} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} \right\} \\
 &= y \left\{ \frac{\partial u}{\partial p} \left(-\frac{x}{y^2} \right) + \frac{\partial u}{\partial q} \left(\frac{1}{z} \right) + 0 \right\} \\
 &= \frac{y}{z} \frac{\partial u}{\partial q} - \frac{x}{y} \frac{\partial u}{\partial p} \quad \text{---- (2)}
 \end{aligned}$$

$$\begin{aligned}
 z \frac{\partial u}{\partial z} &= z \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial z} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} \right\} \\
 &= z \left\{ 0 + \frac{\partial u}{\partial q} \left(-\frac{y}{z^2} \right) + \frac{\partial u}{\partial r} \left(\frac{1}{x} \right) \right\} \\
 &= \frac{z}{x} \frac{\partial u}{\partial r} - \frac{y}{z} \frac{\partial u}{\partial q} \quad \text{----- (3)}
 \end{aligned}$$

(1) + (2) + (3) gives,

$$\begin{aligned}
 x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} &= \frac{x}{y} \frac{\partial u}{\partial p} - \frac{z}{x} \frac{\partial u}{\partial r} + \frac{y}{z} \frac{\partial u}{\partial q} - \frac{x}{y} \frac{\partial u}{\partial p} + \frac{z}{x} \frac{\partial u}{\partial r} - \frac{y}{z} \frac{\partial u}{\partial q} \\
 &= 0
 \end{aligned}$$

6. If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right)$ then prove that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

$p = \frac{y-x}{xy} = \frac{1}{x} - \frac{1}{y}$	$q = \frac{z-x}{zx} = \frac{1}{x} - \frac{1}{z}$
$\frac{\partial p}{\partial x} = -\frac{1}{x^2}$	$\frac{\partial q}{\partial x} = -\frac{1}{x^2}$
$\frac{\partial p}{\partial y} = \frac{1}{y^2}$	$\frac{\partial q}{\partial y} = 0$
$\frac{\partial p}{\partial z} = 0$	$\frac{\partial q}{\partial z} = \frac{1}{z^2}$

Therefore,

$$\begin{aligned}
 x^2 \frac{\partial u}{\partial x} &= x^2 \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} \right\} \\
 &= x^2 \left\{ \frac{\partial u}{\partial p} \left(-\frac{1}{x^2} \right) + \frac{\partial u}{\partial q} \left(-\frac{1}{x^2} \right) \right\} \\
 &= -\frac{\partial u}{\partial p} - \frac{\partial u}{\partial q} \quad \text{---- (1)}
 \end{aligned}$$

$$\begin{aligned}
 y^2 \frac{\partial u}{\partial y} &= y^2 \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial y} \right\} \\
 &= y^2 \left\{ \frac{\partial u}{\partial p} \left(\frac{1}{y^2} \right) + 0 \right\} \\
 &= \frac{\partial u}{\partial p} \quad \text{---- (2)}
 \end{aligned}$$

$$\begin{aligned}
 z^2 \frac{\partial u}{\partial z} &= z^2 \left\{ \frac{\partial u}{\partial p} \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial z} \right\} \\
 &= z^2 \left\{ 0 + \frac{\partial u}{\partial q} \left(\frac{1}{z^2} \right) \right\} \\
 &= \frac{\partial u}{\partial q} \quad \text{---- (3)}
 \end{aligned}$$

(1) + (2) + (3) gives,

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = -\frac{\partial u}{\partial p} - \frac{\partial u}{\partial q} + \frac{\partial u}{\partial p} + \frac{\partial u}{\partial q} = 0$$

7. If $z = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$ then P.T. $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2$

$x = r \cos \theta$	$y = r \sin \theta$
$\frac{\partial x}{\partial r} = \cos \theta = \frac{x}{r}$	$\frac{\partial y}{\partial r} = \sin \theta = \frac{y}{r}$
$\frac{\partial x}{\partial \theta} = -r \sin \theta = -y$	$\frac{\partial y}{\partial \theta} = r \cos \theta = x$

$$\begin{aligned} \left(\frac{\partial z}{\partial r}\right)^2 &= \left(\frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r}\right)^2 \\ &= \left\{\frac{\partial z}{\partial x} \left(\frac{x}{r}\right) + \frac{\partial z}{\partial y} \left(\frac{y}{r}\right)\right\}^2 \\ &= \frac{1}{r^2} \left\{x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}\right\}^2 \quad \text{----- (1)} \end{aligned}$$

$$\begin{aligned} \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2 &= \frac{1}{r^2} \left\{\frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta}\right\}^2 \\ &= \frac{1}{r^2} \left\{-y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y}\right\}^2 \quad \text{----- (2)} \end{aligned}$$

(1) + (2) gives,

$$\begin{aligned} \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2 &= \frac{1}{r^2} \left\{(x^2 + y^2) \left(\frac{\partial z}{\partial x}\right)^2 + (x^2 + y^2) \left(\frac{\partial z}{\partial y}\right)^2\right\} \\ &= \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \quad [\because x^2 + y^2 = r^2] \end{aligned}$$

Home work:

8. If $z = f(u, v)$, $u = x^2 - y^2$, $v = 2xy$ then prove that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(x^2 + y^2) \left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$$

9. If $u = f\left(xz, \frac{y}{z}\right)$ then prove that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} - z \frac{\partial u}{\partial z} = 0$.

10. If $u = f\left(\frac{x}{z}, \frac{y}{z}\right)$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.

2.6 Jacobians

Introduction:

- ❖ If u and v are functions of two independent variables x and y then

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \quad (\text{OR}) \quad J\left(\frac{u,v}{x,y}\right) = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

- ❖ If u, v and w are functions of three independent variables x, y and z then

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} \quad (\text{OR}) \quad J\left(\frac{u,v,w}{x,y,z}\right) = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

Problems:

1. If $x = r \cos \theta$, $y = r \sin \theta$ then prove that $\frac{\partial(x,y)}{\partial(r,\theta)} = r$.

$x = r \cos \theta$	$y = r \sin \theta$
$\frac{\partial x}{\partial r} = \cos \theta$	$\frac{\partial y}{\partial r} = \sin \theta$
$\frac{\partial x}{\partial \theta} = -r \sin \theta$	$\frac{\partial y}{\partial \theta} = r \cos \theta$

$$\begin{aligned} \text{Therefore, } \frac{\partial(x,y)}{\partial(r,\theta)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{vmatrix} \\ &= \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r \end{aligned}$$

2. If $x = u(1 - v)$, $y = uv$ then find $\frac{\partial(x,y)}{\partial(u,v)}$.

$x = u - uv$	$y = uv$
$\frac{\partial x}{\partial u} = 1 - v$	$\frac{\partial y}{\partial u} = v$
$\frac{\partial x}{\partial v} = -u$	$\frac{\partial y}{\partial v} = u$

$$\begin{aligned} \text{Therefore, } \frac{\partial(x,y)}{\partial(u,v)} &= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} \\ &= \begin{vmatrix} 1 - v & v \\ -u & u \end{vmatrix} = u - uv + uv = u \end{aligned}$$

3. If $x = u(1 - v)$, $y = uv$ then find $\frac{\partial(u,v)}{\partial(x,y)}$.

$u = x + y$	$v = \frac{y}{x+y}$
$\frac{\partial u}{\partial x} = 1$	$\frac{\partial v}{\partial x} = -\frac{y}{(x+y)^2}$
$\frac{\partial u}{\partial y} = 1$	$\frac{\partial v}{\partial y} = \frac{x+y-y}{(x+y)^2} = \frac{x}{(x+y)^2}$

$$\begin{aligned}
 \text{Therefore, } \frac{\partial(u,v)}{\partial(x,y)} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} \\
 &= \begin{vmatrix} 1 & -\frac{y}{(x+y)^2} \\ 1 & \frac{x}{(x+y)^2} \end{vmatrix} \\
 &= \frac{x}{(x+y)^2} + \frac{y}{(x+y)^2} \\
 &= \frac{1}{x+y}
 \end{aligned}$$

4. If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$ then prove that u, v and w are functionally dependent.

$u = x^2 + y^2 + z^2$	$v = xy + yz + zx$	$w = x + y + z$
$\frac{\partial u}{\partial x} = 2x$	$\frac{\partial v}{\partial x} = y + z$	$\frac{\partial w}{\partial x} = 1$
$\frac{\partial u}{\partial y} = 2y$	$\frac{\partial v}{\partial y} = x + z$	$\frac{\partial w}{\partial y} = 1$
$\frac{\partial u}{\partial z} = 2z$	$\frac{\partial v}{\partial z} = x + y$	$\frac{\partial w}{\partial z} = 1$

$$\begin{aligned}
 \frac{\partial(u,v,w)}{\partial(x,y,z)} &= \begin{vmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{vmatrix} \\
 &= \begin{vmatrix} 2x & y+z & 1 \\ 2y & z+x & 1 \\ 2z & x+y & 1 \end{vmatrix} \\
 &= 2 \begin{vmatrix} x & y+z & 1 \\ y-x & x-y & 0 \\ z-x & x-z & 0 \end{vmatrix} \\
 &= 2(x-y)(x-z) \begin{vmatrix} x & y+z & 1 \\ -1 & 1 & 0 \\ -1 & 1 & 0 \end{vmatrix} \\
 &= 0
 \end{aligned}$$

Therefore, u, v and w are functionally dependent.

5. If $u = x^2 + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$, evaluate $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at $(1, -1, 0)$.
(May 22)

$u = x^2 + 3y^2 - z^3$	$v = 4x^2yz$	$w = 2z^2 - xy$
$\frac{\partial u}{\partial x} = 2x$	$\frac{\partial v}{\partial x} = 8xyz$	$\frac{\partial w}{\partial x} = -y$
$\frac{\partial u}{\partial y} = 6y$	$\frac{\partial v}{\partial y} = 4x^2z$	$\frac{\partial w}{\partial y} = -x$
$\frac{\partial u}{\partial z} = -3z^2$	$\frac{\partial v}{\partial z} = 4x^2y$	$\frac{\partial w}{\partial z} = 4z$

$$\begin{aligned}\frac{\partial(u,v,w)}{\partial(x,y,z)} &= \begin{vmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{vmatrix} \\ &= \begin{vmatrix} 2x & 8xyz & -y \\ 6y & 4x^2z & -x \\ -3z^2 & 4x^2y & 4z \end{vmatrix} \\ \text{At } (1, -1, 0), \\ &= \begin{vmatrix} 2 & 0 & 1 \\ -6 & 0 & -1 \\ 0 & -4 & 0 \end{vmatrix} = 20\end{aligned}$$

6. If $u = \frac{xy}{z}$, $v = \frac{yz}{x}$, $w = \frac{zx}{y}$ find $J\left(\frac{u,v,w}{x,y,z}\right)$ [Jan 17]

$u = \frac{xy}{z}$	$v = \frac{yz}{x}$	$w = \frac{zx}{y}$
$\frac{\partial u}{\partial x} = \frac{y}{z}$	$\frac{\partial v}{\partial x} = -\frac{yz}{x^2}$	$\frac{\partial w}{\partial x} = \frac{z}{y}$
$\frac{\partial u}{\partial y} = \frac{x}{z}$	$\frac{\partial v}{\partial y} = \frac{z}{x}$	$\frac{\partial w}{\partial y} = -\frac{zx}{y^2}$
$\frac{\partial u}{\partial z} = -\frac{xy}{z^2}$	$\frac{\partial v}{\partial z} = \frac{y}{x}$	$\frac{\partial w}{\partial z} = \frac{x}{y}$

$$\begin{aligned}\frac{\partial(u,v,w)}{\partial(x,y,z)} &= \begin{vmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{vmatrix} \\ &= \begin{vmatrix} \frac{y}{z} & -\frac{yz}{x^2} & \frac{z}{y} \\ \frac{x}{z} & \frac{z}{x} & -\frac{zx}{y^2} \\ -\frac{xy}{z^2} & \frac{y}{x} & \frac{x}{y} \end{vmatrix} \\ &= \frac{1}{x^2y^2z^2} \begin{vmatrix} yz & -yz & yz \\ zx & -zx & -zx \\ -xy & xy & xy \end{vmatrix} \\ &= \frac{x^2y^2z^2}{x^2y^2z^2} \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = 4\end{aligned}$$

7. If $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$ find $J \left(\frac{x,y,z}{r,\theta,\phi} \right)$ [July 16]

$x = r \sin \theta \cos \phi$	$y = r \sin \theta \sin \phi$	$z = r \cos \theta$
$\frac{\partial x}{\partial r} = \sin \theta \cos \phi$	$\frac{\partial y}{\partial r} = \sin \theta \sin \phi$	$\frac{\partial z}{\partial r} = \cos \theta$
$\frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi$	$\frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi$	$\frac{\partial z}{\partial \theta} = -r \sin \theta$
$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi$	$\frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi$	$\frac{\partial z}{\partial \phi} = 0$

$$\begin{aligned}
 \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} &= \begin{vmatrix} x_r & y_r & z_r \\ x_\theta & y_\theta & z_\theta \\ x_\phi & y_\phi & z_\phi \end{vmatrix} \\
 &= \begin{vmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ r \cos \theta \cos \phi & r \cos \theta \sin \phi & 0 \\ -r \sin \theta \sin \phi & r \sin \theta \cos \phi & -r \sin \theta \end{vmatrix} \\
 &= r^2 \sin \theta \begin{vmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{vmatrix} \\
 &= r^2 \sin \theta \{ \cos \theta (\cos \theta) + \sin \theta (\sin \theta) - 1(0) \} \\
 &= r^2 \sin \theta
 \end{aligned}$$

8. If $u = x + y + z, uv = y + z, uvw = z$ find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$. [Jan 16,

$x = u - uv$	$y = uv - uvw$	$z = uvw$
$\frac{\partial x}{\partial u} = 1 - v$	$\frac{\partial y}{\partial u} = v - vw$	$\frac{\partial z}{\partial u} = vw$
$\frac{\partial x}{\partial v} = -u$	$\frac{\partial y}{\partial v} = u - uw$	$\frac{\partial z}{\partial v} = uw$
$\frac{\partial x}{\partial w} = 0$	$\frac{\partial y}{\partial w} = -uv$	$\frac{\partial z}{\partial w} = uv$

$$\begin{aligned}
 \frac{\partial(x,y,z)}{\partial(u,v,w)} &= \begin{vmatrix} x_u & y_u & z_u \\ x_v & y_v & z_v \\ x_w & y_w & z_w \end{vmatrix} \\
 &= \begin{vmatrix} 1-v & v-vw & vw \\ -u & u-uw & uw \\ 0 & -uv & uv \end{vmatrix} \\
 &= u^2 v \begin{vmatrix} 1-v & v-vw & vw \\ -1 & 1-w & w \\ 0 & -1 & 1 \end{vmatrix} \\
 &= u^2 v \begin{vmatrix} 1-v & v & vw \\ -1 & 1 & w \\ 0 & 0 & 1 \end{vmatrix}, c_2 \rightarrow c_2 + c_3 \\
 &= u^2 v (1 - v + v) \\
 &= u^2 v
 \end{aligned}$$

9. If $x = e^u \cos v, y = e^u \sin v$ then find $\frac{\partial(x,y)}{\partial(u,v)}$.

$x = e^u \cos v$	$y = e^u \sin v$
$\frac{\partial x}{\partial u} = e^u \cos v$	$\frac{\partial y}{\partial u} = e^u \sin v$
$\frac{\partial x}{\partial v} = -e^u \sin v$	$\frac{\partial y}{\partial v} = e^u \cos v$

$$\begin{aligned}
 \text{Therefore, } \frac{\partial(x,y)}{\partial(u,v)} &= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} \\
 &= \begin{vmatrix} e^u \cos v & e^u \sin v \\ -e^u \sin v & e^u \cos v \end{vmatrix} \\
 &= e^{2u} \begin{vmatrix} \cos v & \sin v \\ -\sin v & \cos v \end{vmatrix} \\
 &= e^{2u}
 \end{aligned}$$

10. If $u = x^2 - y^2, v = 2xy$ and $x = r \cos \theta, y = r \sin \theta$ find $\frac{\partial(u,v)}{\partial(r,\theta)}$

$u = x^2 - y^2$	$v = 2xy$	$x = r \cos \theta$	$y = r \sin \theta$
$\frac{\partial u}{\partial x} = 2x$	$\frac{\partial v}{\partial x} = 2y$	$\frac{\partial x}{\partial r} = \cos \theta$	$\frac{\partial y}{\partial r} = \sin \theta$
$\frac{\partial u}{\partial y} = -2y$	$\frac{\partial v}{\partial y} = 2x$	$\frac{\partial x}{\partial \theta} = -r \sin \theta$	$\frac{\partial y}{\partial \theta} = r \cos \theta$

$$\begin{aligned}
 \frac{\partial(u,v)}{\partial(x,y)} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} & \frac{\partial(x,y)}{\partial(r,\theta)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{vmatrix} \\
 &= \begin{vmatrix} 2x & 2y \\ -2y & 2x \end{vmatrix} & &= \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix} \\
 &= 4(x^2 + y^2) = 4r^2 & &= r \cos^2 \theta + r \sin^2 \theta = r
 \end{aligned}$$

$$\text{Therefore, } \frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,\theta)} = 4r^2 \times r = 4r^3$$

Home work:

11. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$ find $\frac{\partial(x,y)}{\partial(u,v)}$. Ans: 0

12. If $x = \rho \cos \phi, y = \rho \sin \phi, z = z$ show that $\frac{\partial(x,y,z)}{\partial(\rho,\phi,z)} = \rho$

2.8 Maxima and minima for a function of two variables

Introduction:

❖ **Extreme point:**

A point at which $f(x, y)$ attains maximum or minimum.

❖ **Saddle point:**

A point at which $f(x, y)$ attains neither max. nor minimum.

❖ **Necessary conditions:**

The necessary conditions for $f(x, y)$ to have a max. or min. at (a, b) are

$$\left(\frac{\partial f}{\partial x}\right)_{(a,b)} = 0 \text{ and } \left(\frac{\partial f}{\partial y}\right)_{(a,b)} = 0 .$$

❖ **Notation:**

$$p = \frac{\partial f}{\partial x} , \quad q = \frac{\partial f}{\partial y} , \quad r = \frac{\partial^2 f}{\partial x^2} , \quad t = \frac{\partial^2 f}{\partial y^2} , \quad s = \frac{\partial^2 f}{\partial x \partial y}$$

❖ **Sufficient conditions:**

The sufficient conditions for $f(x, y)$ to have

(i) Max. at (a, b) is that $rt - s^2 > 0$ and $r < 0$.

(ii) Min. at (a, b) is that $rt - s^2 > 0$ and $r > 0$.

Working rule:

❖ Find critical points by solving $p = 0$ and $q = 0$.

❖ Find $rt - s^2$ and r at each critical point.

❖ Write the conclusion using the following table:

At (a, b) , if	$f(x, y)$ attains
$rt - s^2 > 0, r < 0$	Maximum
$rt - s^2 > 0, r > 0$	Minimum
$rt - s^2 < 0$	Neither maximum nor minimum
$rt - s^2 = 0$	Saddle point

1. Show that $f(x, y) = xy(a - x - y)$, $a > 0$ is maximum at the point $\left(\frac{a}{3}, \frac{a}{3}\right)$.

$$\frac{\partial f}{\partial x} = y(a - x - y) - xy = y(a - 2x - y)$$

$$\frac{\partial f}{\partial y} = x(a - x - y) - xy = x(a - x - 2y)$$

$$\text{At } \left(\frac{a}{3}, \frac{a}{3}\right), p = 0 \text{ and } q = 0.$$

Therefore, $\left(\frac{a}{3}, \frac{a}{3}\right)$ is an extremum point.

$$r = \frac{\partial^2 f}{\partial x^2} = -2y$$

$$t = \frac{\partial^2 f}{\partial y^2} = -2x$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = a - 2x - 2y$$

$$rt - s^2 = 4xy - (a - 2x - 2y)^2$$

$$\text{At } \left(\frac{a}{3}, \frac{a}{3}\right), r < 0 \text{ and } rt - s^2 = \frac{4a^2}{9} - \frac{a^2}{9} = \frac{a^2}{3}$$

2. Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3axy$.

$$f(x, y) = x^3 + y^3 - 3axy$$

$$\frac{\partial f}{\partial x} = 3x^2 - 3ay, \quad \frac{\partial f}{\partial y} = 3y^2 - 3ax$$

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 6y, \quad \frac{\partial^2 f}{\partial x \partial y} = -3a$$

Step:1 Find critical points

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow x^2 = ay \text{ ---- (1)}$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow y^2 = ax \text{ ---- (2)}$$

$$(1) \times x - (2) \times y \Rightarrow x^3 = y^3 \Rightarrow x = y$$

Put $y = x$ in (1). We get Critical points $(0, 0)$, (a, a) .

Step:2 Find $rt - s^2$ at each critical point

$$r = 6x, \quad t = 6y, \quad s = -3a, \quad rt - s^2 = 36xy - 9a^2$$

Critical points	$rt - s^2 = 36xy - 9a^2$	$r = 6x$	Remark
$(0, 0)$	0	----	Saddle point
(a, a)	$36a^2 - 9a^2$, Positive.	Positive	Minimum

Step:3 Conclusion

$f(x, y)$ attains minimum at (a, a) .

$$\text{Minimum value} = f(a, a) = a^3 + a^3 - 3a^3 = -a^3.$$

3. Find the extreme values of the function $f(x, y) = x^3 + 3x^2 + 4xy + y^2$.

$$f(x, y) = x^3 + 3x^2 + 4xy + y^2$$

$$\frac{\partial f}{\partial x} = 3x^2 + 6x + 4y, \quad \frac{\partial f}{\partial y} = 4x + 2y$$

$$\frac{\partial^2 f}{\partial x^2} = 6x + 6, \quad \frac{\partial^2 f}{\partial y^2} = 2, \quad \frac{\partial^2 f}{\partial x \partial y} = 4$$

Step:1 Find critical points

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow 3x^2 + 6x + 4y = 0 \text{---- (1)}$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow 4x + 2y = 0 \Rightarrow y = -2x \text{---- (2)}$$

$$\text{Substitute (2) in (1)} \Rightarrow 3x^2 + 6x - 8x = 0 \Rightarrow x = 0, \frac{2}{3}$$

$$\text{In (2), } x = 0 \Rightarrow y = 0 \text{ and } x = \frac{2}{3} \Rightarrow y = -\frac{4}{3}$$

Therefore, Critical points are $(0,0)$, $(\frac{2}{3}, -\frac{4}{3})$.

Step:2 Find $rt - s^2$ at each critical point

$$r = 6x + 6, \quad t = 2, \quad s = 4$$

$$rt - s^2 = 12x + 12 - 16$$

Critical points	$rt - s^2 = 12x - 4$	$r = 6x + 6$	Remark
$(0,0)$	-4 , Negative	----	Neither max. nor min.
$(\frac{2}{3}, -\frac{4}{3})$	$8 - 4$, Positive.	Positive	Minimum

Step:3 Conclusion

$f(x, y)$ attains minimum at $(\frac{2}{3}, -\frac{4}{3})$.

$$\begin{aligned} \text{Minimum value} &= f\left(\frac{2}{3}, -\frac{4}{3}\right) \\ &= \left(\frac{2}{3}\right)^3 + 3\left(\frac{2}{3}\right)^2 + 4\left(\frac{2}{3}\right)\left(-\frac{4}{3}\right) + \left(-\frac{4}{3}\right)^2 \\ &= -\frac{4}{27} \end{aligned}$$

4. Find the extreme values of the function $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$.

$$f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$$

$$\frac{\partial f}{\partial x} = 4x^3 - 4x + 4y, \quad \frac{\partial f}{\partial y} = 4y^3 + 4x - 4y$$

$$\frac{\partial^2 f}{\partial x^2} = 12x^2 - 4, \quad \frac{\partial^2 f}{\partial y^2} = 12y^2 - 4, \quad \frac{\partial^2 f}{\partial x \partial y} = 4$$

Step:1 Find critical points

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow 4x^3 - 4x + 4y = 0 \Rightarrow x^3 - x + y = 0 \text{ ---- (1)}$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow 4y^3 + 4x - 4y = 0 \Rightarrow y^3 - y + x = 0 \text{ ---- (2)}$$

$$(1) + (2) \Rightarrow y = -x$$

$$\text{In (2), } y = -x \Rightarrow -x^3 + x + x = 0, \quad x(2 - x^2) = 0$$

Therefore, Critical points are $(0,0)$, $(\sqrt{2}, -\sqrt{2})$, $(-\sqrt{2}, \sqrt{2})$

Step:2 Find $rt - s^2$ at each critical point

$$r = 12x^2 - 4, \quad t = 12y^2 - 4, \quad s = 4$$

$$rt - s^2 = (12x^2 - 4)(12y^2 - 4) - 16$$

Critical points	$rt - s^2 =$ $(12x^2 - 4)(12y^2 - 4) - 16$	$r =$ $12x^2 - 4$	Remark
$(0, 0)$	0	----	Doubtful
$(\sqrt{2}, -\sqrt{2})$	$400 - 16$, Positive	Positive	Minimum
$(-\sqrt{2}, \sqrt{2})$	$400 - 16$, Positive	Positive	Minimum

Step:3 Conclusion

$f(x, y)$ attains minimum at $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$.

$$\text{Minimum value} = f(\sqrt{2}, -\sqrt{2}) = f(-\sqrt{2}, \sqrt{2}) = 4 + 4 - 4 - 8 - 4 = -8$$

5. Find the extreme values of the function $f(x, y) = x^3y^2(1 - x - y)$.

$$f(x, y) = x^3y^2 - x^4y^2 - x^3y^3$$

$$\frac{\partial f}{\partial x} = 3x^2y^2 - 4x^3y^2 - 3x^2y^3, \quad \frac{\partial f}{\partial y} = 2x^3y - 2x^4y - 3x^3y^2$$

$$\frac{\partial^2 f}{\partial x^2} = 6xy^2 - 12x^2y^2 - 6xy^3, \quad \frac{\partial^2 f}{\partial y^2} = 2x^3 - 2x^4 - 6x^3y,$$

$$\frac{\partial^2 f}{\partial x \partial y} = 6x^2y - 8x^3y - 9x^2y^2$$

Step:1 Find critical points

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow 3x^2y^2 - 4x^3y^2 - 3x^2y^3 = 0 \Rightarrow x^2y^2(3 - 4x - 3y) = 0 \quad \text{---- (1)}$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow 2x^3y - 2x^4y - 3x^3y^2 = 0 \Rightarrow x^3y(2 - 2x - 3y) = 0 \quad \text{---- (2)}$$

$$3 - 4x - 3y = 0 \quad \begin{array}{|c|c|} \hline 0 & 1 \\ \hline 3/4 & 0 \\ \hline \end{array} \quad \text{From (1), } x = 0 \text{ or } \frac{3}{4}, y = 0 \text{ or } 1$$

If $x = 0$, (2) gives $y = 0$ or $2/3$. If $x = 3/4$, (2) gives $y = 0$.

If $y = 0$, (2) gives $x = 0$ or 1 . If $y = 2/3$, (2) gives $x = 0$.

Therefore, Critical points are $(0, 0), (0, \frac{2}{3}), (\frac{3}{4}, 0), (\frac{1}{2}, \frac{1}{3}), (1, 0)$.

Step:2 Find $rt - s^2$ at each critical point

	$(0, 0)$	$(0, 2/3)$	$(3/4, 0)$	$(1/2, 1/3)$	$(1, 0)$
$r = 6xy^2(1 - 2x - y)$	0	0	0	-1/9	0
$t = 2x^3(1 - x - 3y)$	0	0	1/128	-1/8	0
$s = x^2y(6 - 8x - 9y)$	0	0	0	-1/12	0

Critical points	$rt - s^2$	r	Remark
$(0, 0)$	0	0	Saddle point
$(0, 2/3)$	0	0	Saddle point
$(3/4, 0)$	0	0	Saddle point
$(1/2, 1/3)$	Positive	Negative	Maximum
$(1, 0)$	0	0	Saddle point

Step:3 Conclusion

$f(x, y)$ attains maximum at $(\frac{1}{2}, \frac{1}{3})$. Minimum value $= f(\frac{1}{2}, \frac{1}{3}) = \frac{1}{432}$

6. Find the extreme values of the function $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$.

$$f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$$

$$\frac{\partial f}{\partial x} = 3x^2 + 3y^2 - 6x, \quad \frac{\partial f}{\partial y} = 6xy - 6y$$

$$\frac{\partial^2 f}{\partial x^2} = 6x - 6, \quad \frac{\partial^2 f}{\partial y^2} = 6x - 6, \quad \frac{\partial^2 f}{\partial x \partial y} = 6y$$

Step:1 Find critical points

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow 3x^2 + 3y^2 - 6x = 0 \Rightarrow x^2 + y^2 - 2x = 0 \text{ ---- (1)}$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow 6xy - 6y = 0 \Rightarrow y = 0, x = 1. \text{ ---- (2)}$$

Substitute $y = 0$ in (1) $\Rightarrow x = 0$ or 2 .

Substitute $x = 1$ in (1) $\Rightarrow y = -1$ or 1 .

Therefore, Critical points are $(0, 0), (2, 0), (1, -1), (1, 1)$.

Step:2 Find $rt - s^2$ at each critical point

$$r = 6x - 6, \quad t = 6x - 6, \quad s = 6y$$

$$rt - s^2 = 36(x - 1)^2 - 36y^2$$

Critical points	$rt - s^2$	$r = 6x - 6$	Remark
$(0, 0)$	Positive	-6	Maximum
$(2, 0)$	Positive	6	Minimum
$(1, 1)$	Negative	-----	
$(1, -1)$	Negative	-----	

Step:3 Conclusion

(i) $f(x, y)$ attains maximum at $(0, 0)$. Maximum value $= f(0, 0) = 4$

(ii) $f(x, y)$ attains minimum at $(2, 0)$. Minimum value $= f(2, 0) = 0$

7. Find the extreme values of $f(x, y) = x^3 + y^3 - 63x - 63y + 12xy$.

$$f(x, y) = x^3 + y^3 - 63x - 63y + 12xy$$

$$\frac{\partial f}{\partial x} = 3x^2 - 63 + 12y, \quad \frac{\partial f}{\partial y} = 3y^2 - 63 + 12x$$

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 6y, \quad \frac{\partial^2 f}{\partial x \partial y} = 12$$

Step:1 Find critical points

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow 3x^2 + 12y = 63 \Rightarrow x^2 + 4y = 21 \text{ ---- (1)}$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow 3y^2 + 12x = 63 \Rightarrow y^2 + 4x = 21. \text{ ---- (2)}$$

$$(1) - (2) \Rightarrow (x^2 - y^2) - 4(x - y) = 0 \Rightarrow (x - y)(x + y - 4) = 0$$

Therefore, $x = y$ or $y = 4 - x$

If $x = y$, (1) becomes $x^2 + 4x - 21 = 0 \Rightarrow x = 3, -7$

If $y = 4 - x$, (1) becomes $x^2 - 4x - 5 = 0 \Rightarrow x = 5, -1$

Therefore, Critical points are $(3, 3), (-7, -7), (5, -1), (-1, 5)$.

Step:2 Find $rt - s^2$ at each critical point

$$r = 6x, \quad t = 6y, \quad s = 12$$

$$rt - s^2 = 36xy - 144$$

Critical points	$rt - s^2$	$r = 6x$	Remark
$(3, 3)$	Positive	18	Minimum
$(-7, -7)$	Positive	-42	Maximum
$(-1, 5)$	Negative	-----	
$(5, -1)$	Negative	-----	

Step:3 Conclusion

(i) $f(x, y)$ attains minimum at $(3, 3)$. Minimum value $= f(3, 3) = -216$

(ii) $f(x, y)$ attains maximum at $(-7, -7)$. Maximum value $= f(-7, -7) = 784$

Home work:

8. Examine the function $f(x, y) = 1 + \sin(x^2 + y^2)$ for extreme values.
Ans: $A = (a, b)$ such that $a^2 + b^2 = \frac{\pi}{2}$ is the max. point. Max. value is 2.
9. Find the extreme values of the function $f(x, y) = \sin x \sin y \sin(x + y)$.
Ans: $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$ is the max. point. Max. value is $\frac{3\sqrt{3}}{8}$.
10. Show that $f(x, y) = x^3 + y^3 - 3x - 12y + 20$ has a maximum value at the point $(-1, -2)$ and a minimum value at the point $(1, 2)$. (May 22)